

Mood-Based Playlist Creator (Prototype) - Documentation

Overview

AI-powered app that detects emotions from facial expressions and recommends personalized music playlists in English, Tamil, and Hindi.

Features

- Real-time emotion detection via webcam
- Multi-language song recommendations (English/Tamil/Hindi)
- Visual emotion analysis charts
- Direct YouTube links to songs

Tech Stack

- **Python** with OpenCV (face detection)
- **DeepFace** (emotion analysis)
- **Matplotlib** (visualization)
- **Google Colab** (cloud execution)

Supported Emotions

1. 😊 Happy
2. 😞 Sad

3. 😡 Angry
4. 😮 Surprise
5. 😨 Fear
6. 🤢 Disgust
7. 😐 Neutral

Quick Start

Installation

```
!pip install deepface opencv-python matplotlib
```

Run Application

```
if __name__ == "__main__":  
    main()
```

How It Works

1. **Capture** - Take photo via webcam
2. **Analyze** - AI detects facial emotions
3. **Recommend** - Get personalized playlists
4. **Enjoy** - Click YouTube links to listen

Usage Tips

- Good lighting ensures better detection
- Show clear facial expressions
- Avoid sunglasses/hats

- Stay still during capture

Languages Available

- **English** - International hits
- **Tamil** - South Indian music
- **Hindi** - Bollywood/pop songs

Performance

- Processing time: 2-5 seconds
- Accuracy: ~85% emotion detection
- Works on any device with webcam

Note: For educational/entertainment purposes. Accuracy may vary based on lighting and image quality.

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